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Answer First, Reason Later: Commitment Order in Diffusion LLMs

Unconstrained Masked Decoding Commits Final Answers Before Reasoning Completes, Collapsing to Answer-Only Outputs on Up to 90% of Problems

By Jewon Yeom, Jaewon Sok, Seonghyeon Park, Jeongjae Park, Hwiyeong Lee, Taesup Kim

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Masked diffusion language models promise order-free generation as a core advantage over autoregressive decoding. A new study from Seoul National University shatters that assumption for reasoning tasks: logging every token commitment in LLaDA-8B on GSM8K reveals that unconstrained decoding commits the final answer at only 15–24% of the trajectory, before half the reasoning region is written, collapsing to answer-only outputs on up to 90% of problems as canvas size grows.

AT A GLANCE

Problem: Diffusion LLMs generate tokens in arbitrary order, but reasoning tasks require intermediate steps to precede the final answer — an ordering constraint the model violates.

Why it matters: dLLMs are increasingly proposed as reasoning engines; if their decoding procedure systematically skips reasoning, the architecture is unreliable for this use case.

Method: Log every token commitment (masked $\rightarrow$ unmasked transition) during LLaDA-8B decoding on GSM8K; analyze commitment position as a fraction of the decoding trajectory.

Result: Pure decoding commits the answer at 15–24% of the trajectory; constrained ordering recovers substantial accuracy without retraining.

Evidence: Collapse to answer-only outputs reaches 90% of problems at large canvas size; commitment trajectory visualizations confirm answer-first ordering.

The Big Picture

Masked diffusion language models (dLLMs) generate text by starting from a fully masked token sequence and iteratively unmasking tokens over multiple diffusion steps. Unlike autoregressive (AR) models, which are constrained

to generate left-to-right, dLLMs may unmask any position in any order. Proponents argue this flexibility enables superior global coherence and parallel generation.

For reasoning tasks, however, this flexibility has a fatal cost. A reasoning trace has inherent logical structure: the intermediate steps should be generated *before* the final answer, because the answer should be justified by preceding computation, not the reverse. When a dLLM commits the answer token before writing the reasoning steps, it is, in effect, guessing — and the reasoning region becomes post-hoc decoration for a decision already made.

Why Existing Approaches Fall Short

Standard dLLM decoding (“pure” diffusion) selects which tokens to unmask at each step based on model confidence, without any constraint on the semantic identity of the tokens being committed. The model may unmask the final answer position early because the answer is high-confidence given the problem statement alone — before the chain-of-thought region provides any additional information.

Prior work on improving dLLM reasoning has focused on training objectives (RLVR, supervised fine-tuning on CoT data) or on post-hoc verification. None has examined the *order* in which tokens are committed during decoding, which this paper identifies as the proximate cause of failure.

Core Mechanism

The authors introduce **commitment order logging**: at each diffusion step, they record which output position transitions from masked to unmasked, forming a complete commitment trajectory. For a generation with T positions and S decoding steps:

- Each step s produces a set $C_s \subseteq [T]$ of newly unmasked positions.
- The answer region $\mathcal{A} \subset [T]$ and reasoning region $\mathcal{R} \subset [T]$ are identified by position in the output template.
- The “answer commitment step” $s^* = \min\{s : C_s \cap \mathcal{A} \neq \emptyset\}$ is the fraction s^*/S of the trajectory at which the final answer is first committed.

The **constrained decoding** fix enforces that positions in $\mathcal{A}$ remain masked until a threshold fraction τ of the trajectory has elapsed, releasing the answer region only after substantial reasoning has been committed.

Technical Formulation

Let $\mathbf{x}_0$ be the target output and $\mathbf{x}_s$ the partially masked sequence at step s , with $\mathbf{x}_S$ fully masked. Pure reverse

diffusion selects tokens to unmask according to:

$$p_{\theta}(\mathbf{x}_{s-1} \mid \mathbf{x}_s) \propto \prod_{i \in [T]} p_{\theta}(x_i \mid \mathbf{x}_s)^{\mathbb{1}_{[x_s^{(i)} = \text{[MASK]}]}}$$

Commitment at step s selects the top- k unmasked positions by $p_{\theta}(x_i \mid \mathbf{x}_s)$.

Constrained decoding modifies the selection by adding a region-based gate:

$$p_{\theta}^{\text{con}}(x_i \mid \mathbf{x}_s) = p_{\theta}(x_i \mid \mathbf{x}_s) \cdot \mathbb{1}_{\left[i \notin \mathcal{A} \text{ or } \frac{s}{S} > \tau\right]}$$

where $\tau \in (0, 1)$ is the threshold fraction. No change to model weights is required; the constraint is purely at inference time.

Learning or Inference Procedure

No training is involved. Both pure decoding and constrained decoding are evaluated at inference time on LLaDA-8B, a masked diffusion language model with 8 billion parameters. GSM8K provides 1,319 test-set math word problems as the evaluation benchmark. The commitment threshold τ is tuned on a small held-out validation set.

What the Guarantee Says

No formal guarantee is provided. The empirical claim is that answer-first commitment is the dominant failure mode of pure dLLM decoding on reasoning tasks, and that constraining the answer region during early decoding consistently recovers accuracy. The correlation between canvas size and collapse rate (peaking at 90% answer-only outputs) establishes that the failure is deterministic as generation length grows, not stochastic.

Experimental Findings

- **Commitment timing:** Pure decoding commits the answer at 15–24% of the decoding trajectory on average.
- **Collapse rate:** Up to 90% of problems produce answer-only outputs (zero reasoning steps) as canvas size grows.
- **Constrained decoding:** Enforcing reasoning-before-answer commitment recovers substantial GSM8K accuracy with threshold $\tau = 0.5$.
- **Scale sensitivity:** The gap between constrained and unconstrained decoding widens with output length, confirming scale-dependent failure.

Ablations and Interpretation

Why does pure decoding answer first? The answer token has high local confidence given the problem

statement alone (the model “knows” plausible answer formats). The reasoning region is globally coherent only *after* intermediate steps are committed — but those steps are low-confidence before being written, so the decoding procedure deprioritizes them.

Why does collapse worsen with canvas size? Longer outputs have more reasoning positions that must be committed before the answer is released. As the reasoning region grows, the denominator of the commitment problem grows — making early, spurious answer commitment increasingly likely under unconstrained selection.

Threshold sensitivity: Accuracy peaks around $\tau = 0.4$ – 0.6 with a shallow optimum; extreme values ($\tau \rightarrow 0$ or $\tau \rightarrow 1$) recover the failure modes of pure decoding or fully blocked answer generation respectively.

Reference

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